Angle between two lines

1 Find the angle, θ, between the following lines.

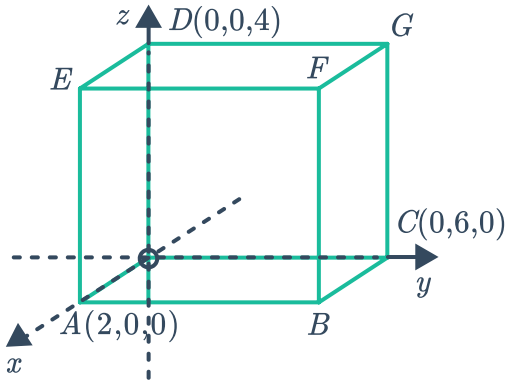
a A line with an angle of inclination 28° and a line with an angle of inclination 79°.

b A line with an angle of inclination 35° and a line with an angle of inclination 167°.

Angle between a line and a plane

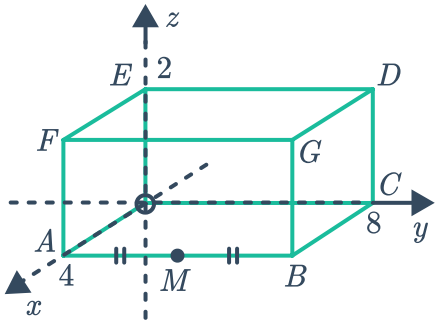
2 Find the angle, correct to one decimal place, between the plane $AOCB$ at the base of the prism, and the following lines:

a AF **b** AG **c** BD



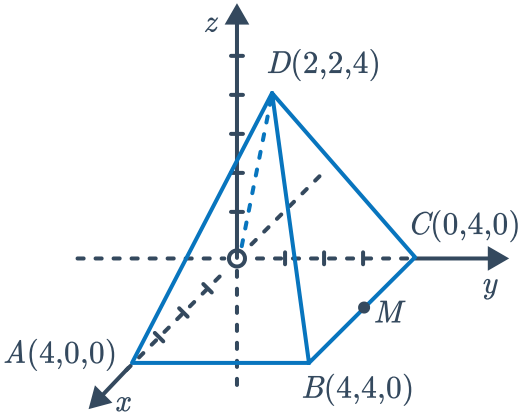
3 Find the angle, correct to one decimal place, between the plane $AOCB$ at the base of the prism, and the following lines:

a AE **b** CF **c** MD



4 Consider the following square pyramid where M is the midpoint of BC:
Find the angle, correct to one decimal place, between the plane $AOCB$ at the base of the pyramid, and the following lines:

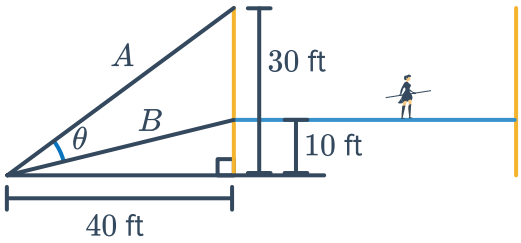
a AD **b** MD



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Application

5 In a circus, a wire, A, is attached to the top of a 30 ft pole. Wire B is used for performers to walk up to the tight wire, 10 ft above the ground. The wires are attached 40 ft from the pole.
Find the angle θ between the wires. Round your answer to two decimal places.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647